archived web as a heuristic problem for a historian. Rightly so, since when born-digital content is stored, a new layer of digitality is generated, the data are reorganized for archival purposes and the user interface strongly influences the ways in which the historian can interact with the 'archive'. This aspect is worth underlining since the amount of born-digital content has increased, and is increasing, exponentially. This also places new expectations on the future archive, or database, of digital content. This archive should be able to preserve the present-day social media debates, online publications, image collections, mobile games and software, user-generated data sets and many other digital remains of the present for future generations to explore. In this work it is crucial to openly articulate the principles of reborn-digital data to offer researchers the possibility to critically assess how this dimension of digitality influences the ways in which the data, in the end, can tell us about the past.

2
Reading and Textuality in Digital History

'Some books are to be tasted, others to be swallowed, and some few to be chewed and digested', wrote Francis Bacon in his classic essay, 'Of studies', published in the late sixteenth century. He continued: 'Some books are to be read only in parts; others to be read but not curiously; and some few to be read wholly, and with diligence and attention.'[1] Bacon's famous words capture the multifaceted experience of reading, its different rhythms and functions, and also the fact that reading is always related to what, in the end, is being read and what are those texts that we encounter. For digital historians, the role of texts and textuality as well as the different practices of reading are essential. This chapter will first discuss reading as a research practice that is, and has been, vital for historians in their effort to understand the past.

Reading is often entertaining and sometimes painful and laborious, but it is also central to our understanding of the world and our digestion of information. Historical research has often been characterized as a realm of books. Historians must engage in two dialogues at the same time; they must acquire knowledge from the sources of the past and converse with other historians, both past and present. Reading is a research practice. The same applies in the case of the digital historian, who simultaneously engages in the centuries-old

traditions of reading while adopting new ways of interpreting sources. The question of reading has been challenged by the expansion of texts that have to be digested and the idea of distant reading, advocated by literary scholars, such as Franco Moretti. The chapter will end by discussing texts as big data, as sources that can be used for different distant reading techniques.

Reading as a research method

In his analysis on the history of reading, James Raven describes reading as a 'notoriously problematic' exercise because the act of reading seems to eschew recording; 'few write down what they do'.[2] Some might write comments in the margins or remarks in their notebooks; others might keep a reading diary. But still, the actual process remains enigmatic. What happens when a historian reads? In his *Going to the Sources: A Guide to Historical Research and Writing*, Anthony Brundage tried to make this process visible and stressed the need to read throughout one's research, not to defer from it but to engage in it right from the start.[3] There are also many kinds of reading habits, or registers of reading, that need to be activated. On the one hand, a scholar must be an effective reader, able to retain and process a lot of information, but s/he must be a critical reader, too, to be able to filter out misleading or irrelevant information. Furthermore, a scholar must track down and verify many details, such as basic background information on the authors of primary sources, and to compare previous works of history to understand the state of the art.

Reading is a method of acquiring information – of 'perusing written or printed matter'[4] – but it has wider ramifications because it involves interpretative work: efforts to identify similarities and differences, continuities and discontinuities, the testing of possible explanations and, ultimately, the drawing of conclusions. Rolf Engelsing, the German historian of reading, saw a conceptual distinction in the perusing of text, noting that extensive reading is needed when a larger overview of the material is required, and sometimes intensive reading is required when it is necessary to delve into the details.[5] In the latter case, reading the text only once is insufficient; one has to read the same texts again and again to understand their underlying meanings and significant structures. Reading is never simply a matter of decoding textual signs. Rather, it involves one's memory of what has been read before, either in the same texts or previous ones. In addition to past experiences, reading also incorporates expectations about the future, since the now of reading continuously produces hypotheses about the possible pathways of forthcoming developments in the work.

Since the 1990s, historians have recurrently used the expression 'close reading' to describe their method of reading.[6] Often, close reading refers to careful, intensive reading during which historians are close to the details while at the same time seeing larger contexts between the lines. The notion of 'close reading' has become widespread in historical studies after the rise of the so-called new cultural history in the 1980s and especially 1990s.[7] As a notion, close reading was borrowed from literary studies, in which its meaning was somewhat different. It was especially advocated by practitioners of the so-called New Criticism after the Second World War.[8] The new critics wanted to defer from cultural, historical and, especially, biographical contextualization and to interpret literary works as self-contained aesthetic objects. The idea behind close reading was to concentrate on the form, literary devices and internal structures employed in autonomous works of art.[9] It can be argued that, since the 1990s, close reading has – at least in the historians' discourse – emphasized careful examination of the form and expressions of the source but, in contrast to New Criticism, aimed at interpreting these structural aspects in the wider framework of cultural and historical change.

The idea of close reading is related to the question of how texts make meaning. To be sure, the interest in textual signification has many backgrounds, but it was particularly inspired by semiotics and its interest in signs and signifiers in the 1960s and 1970s. As an expression, close reading is most commonly used to describe scholarly work on texts, but in the semiotic understanding of culture, reading has meanings other than what can be encountered in written or printed texts in and of themselves. The semiotic turn

emphasized language in a more general way. We have natural languages, like Hindi, Korean and Portuguese, but semioticians interpreted other means of symbolic communication as 'languages', too. They argued, for example, that images can also be seen as texts with their own specific language. Sixteenth-century paintings were based on their own visual motifs, figures, ornaments and so on. Musical language, in turn, could consist of, say, melodies, chords and rhythmic units.[10] The general idea from the point of view of reading has been that we can *read* not only written or printed texts but many kinds of cultural products, and that reading refers to the interpretation and decoding of signs and symbols. The readers just have to know the different registers of this communication in the same way they know the natural language of the book, specifically its vocabulary and syntax.

Reading with machines

Today, in the middle of the rapidly changing media world, the word 'reading' is increasingly used in cases where the reader is not actually a human being but a technological device. There are optical readers used, for example, in scanners to capture visual information and convert it into other formats so it can be further elaborated on computers. A barcode on the back cover of a book includes information about the product, and it can be optically read so that this information can be channelled to another system in order to use it on an invoice at the bookshop. Here, 'reading' refers both to the decoding and transformation of data. Many of the digitization projects discussed in the previous chapter draw on the idea of optical reading. The vast collections of digital corpora have been produced using optical character recognition (OCR), which is the conversion of images into text. Recognition is based on the visual features of characters, and the essential element is that the text it produces is machine-readable, allowing a computerized analysis of the texts which would not be possible in the analogue form. The historical background of OCR is actually related to more conventional reading or the necessity to find new avenues for reading. Already in the early twentieth century, there was a need for technology that

could help to convert printed texts into a form that would not require human eyes. In 1913, Edmund Edward Fournier d'Albe developed his optophone, which scanned text and changed the characters into tones that enabled the blind to hear the letters and thus read without being able to see any of the characters.[11] In subsequent decades, many technologies were developed to enhance the automatic recognition of characters.[12]

The digital humanities are based on the premise that data is machine-readable, meaning that information can be read not only by the researcher but also by a computer or software. An optical reader is only one device in the process of enabling machine-readability. If we define 'reading' as both digesting and organizing information, machines can certainly read. If we further suggest that reading is interpretation as well, it is evident that machines can be taught to analyse and make interpretations of the data they have internalized. To further complicate the issue, in the present situation, we are not only readers ourselves, but we are also objects of reading in new and imaginative ways. The algorithmic 'reading' of our words and deeds is an urgent issue. This means, too, that the cultural historical context of reading has profoundly changed over the last few decades.

Distant reading

The theme of this volume – digital history – is based on the current cultural condition in which we have big data, both digitized and born-digital, in machine-readable forms, which allows for the development of new methodologies for historical research and, unquestionably, new kinds of research settings and historical inquiries. If the nature, quality and extent of the source material available for research have changed, it is natural that the research toolbox must change in tandem. In this situation, the notion of close reading, cultivated extensively by historians and other researchers in the humanities for decades, has been challenged by the idea of *distant reading* in the 2000s. The term 'distant reading' became popular as a polemic commentary on close reading. It was coined by the literary scholar Franco Moretti in his

article 'Conjectures on world literature', published in 2000 in the *New Left Review*. Moretti argued that distant reading allowed us 'to focus on units that are much smaller or much larger than the text: devices, themes, tropes – or genres and systems'.[13] Instead of reading too closely, the reader had to get some distance to be able to see larger features that otherwise would not catch the eye as well as those smaller, subtle elements that gain weight only when they accumulate. In his 2005 book *Graphs, Maps, Trees: Abstract Models for a Literary History*, Moretti wrote that this distance 'is not an obstacle, but a specific form of knowledge: fewer elements, hence a sharper sense of their overall interconnection. Shapes, relations, structures. Forms. Models.'[14]

In his original critique in the *New Left Review*, Moretti provocatively presented close and distant reading as opposites. He even argued that the distance is 'a condition of knowledge'.[15] There are many grounds for this conclusion that will be discussed later in this chapter, as well as arguments that are not necessarily relevant for digital historians since, as I see it, the very ideas of closeness and its opposite, distance, can vary. There are many publications that try to tackle this dichotomy and go beyond the dualistic setting.[16]

The notion of distant reading is today employed frequently in the study of the digital humanities, and it is relevant for digital history too if the source material contains texts *en masse*. It has also meant the return of statistical analyses and quantitative methods after an era of more interpretive approaches in cultural studies. Often, distant reading is presented simply as a way of analysing 'hundreds or thousands of books', or even as something that is more 'objective' than close reading.[17] There are, however, several aspects that must be taken into consideration when interpreting Moretti. First, his approach arises from literary studies, in which there has always been plenty of discussion on different reading methods. As already argued, close reading as a practice was originally presented against the biographical interpretation of literary works, pointing out the importance of analysing the inner structures of texts and delving into the level of words and syntax. In literary studies, many other catchwords for reading have been emphasized. The notion of symptomatic reading, fuelled by psychoanalysis and Marxism, was based

on the assumption that the real meaning of a text lies in what it does not say, which means that the reader has to try to go beyond the surface and find hidden meanings. As an opposite, surface reading was introduced to take the face value of the text seriously and analyse 'the complexity of literary surfaces' that had been 'rendered invisible by symptomatic reading'.[18]

To be sure, Moretti underlines the study of numerous texts through digital methods. But, in applying his thoughts in historical studies, one must be conscious of the fact that his formulations of distant reading were, at least in their first iteration in the *New Left Review*, targeted against the literary canon. His point of departure was challenging the conventional understanding of world literature. Moretti writes:

> What does it mean, studying world literature? How do we do it? I work on West European narrative between 1790 and 1930, and already feel like a charlatan outside of Britain and France. World literature?
>
> Many people have read more and better than I have, of course, but still, we are talking of hundreds of languages and literatures here. Reading 'more' seems hardly to be the solution. Especially because we've just started rediscovering what Margaret Cohen calls the 'great unread'. 'I work on West European narrative, etc. ...' Not really, I work on its canonical fraction, which is not even one per cent of published literature. And again, some people have read more, but the point is that there are thirty thousand nineteenth-century British novels out there, forty, fifty, sixty thousand – no one really knows, no one has read them, no one ever will.[19]

Moretti was clearly inspired by Margaret Cohen's idea of the 'great unread', presented in her 1999 book *The Sentimental Education of the Novel*. With this term, Cohen referred to that huge body of texts that had become unavailable or had been completely forgotten.[20] This created a challenge for research. How can one know what these forgotten books really covered? What did they discuss? And why were they popular in their own time? One researcher could only work on a small segment of this vast collection of 'unread' books.

This observation was in fact already an argument for trying to find new methodologies for studying literature.

Moretti's aim was to advocate the idea of *Weltliteratur*, in the spirit of Goethe and Marx,[21] but with modern methods. As he argued, close reading 'necessarily depends on an extremely small canon', since it is not possible to closely analyse as many texts as the understanding of world literature would require. Therefore, new methods are needed, and the researcher should 'learn how *not* to read'. Furthermore, Moretti poignantly points out that 'The United States is the country of close reading, so I don't expect this idea to be particularly popular.'[22]

It is obvious that Moretti's provocation was addressed in particular to literary scholars who analysed one single author or work and thus strengthened the existing canon. The choices of the researcher were not impartial but participated in constructing the notion of literature. It is important to note this when applying Moretti's ideas on distant reading in other fields of the humanities. Additionally, it should not be taken for granted that Moretti's representation of literary studies is doing justice to the array of different approaches and trends in his own field. Regardless, in historical studies, the relationship of the researcher to texts and textuality is clearly different from Moretti's image, since historians often tend to incorporate as much material as possible into their investigations, instead of concentrating on only a few texts.

Remarkably, Moretti makes a reference to historians, too. He quotes Marc Bloch's slogan 'Years of analysis for a day of synthesis'.[23] and refers to the overarching approaches of Fernand Braudel and Immanuel Wallerstein. He points out that, in the end, the synthesized parts of, for example, Wallerstein's works are very condensed. The author crystallized his 'years of analysis' into 'one third of a page'.[24] This reveals Moretti's aim: he is advocating efforts to find new ways for creating abstractions based on massive amounts of data. For him, visualizations serve as a means to synthesize those observations made through computational methods. This is what he further developed in his book *Graphs, Maps, Trees: Abstract Models for a Literary History*.[25]

Distant reading underlines the idea of distance: the researcher should not be too close to the text: '[T]he more

ambitious the project, the greater must the distance be.'[26] In the context of historical studies, this view can certainly be contested. It is essential to draw on big data, now available for research, and explore novel ways of observing this material from a distance, but at the same time, it is equally important to return *from* the distance back to the close encounter of individual texts and consider how the bigger picture might change our view of the details. This continuous motion between distance and closeness has been emphasized by many researchers since Moretti.[27] Scott Weingart has pointed out the moral aspect of the distance by arguing: 'When you zoom out far enough, everything looks the same.' Therefore, it is necessary for researchers in the humanities 'to bring our close reading perspectives to the distant view'.[28] Frédéric Clavert, in turn, has suggested that historians should 'be able to perform a double reading (*une double lecture*)', to work in the archives and keep close relationships with sources via a human reading; at the same time, however, they should be able to look at everything from afar.[29] Sometimes distancing from distant reading is necessary; it can give new insights into close reading, and vice versa. Tim Hitchcock, one of the pioneers of digital history and a developer of Old Bailey Online, a repository of London's Central Criminal Court, 1674–1913, has highlighted the importance of being able to scale between the macro and micro levels and emphasized the potential of 'computer-assisted close reading'.[30]

In distant reading, the word 'distant' has a twofold meaning: it refers to the effort of trying to refrain from delving into detail and keeping away from the textual surface of the source, and it also insinuates that the research data are so vast that they simply cannot be viewed *but* from a distance. Big data requires a distant approach since close reading is not possible – let us think back to those hundreds of millions of books that have been digitized. It would not be humanly possible to read them all, and they would perhaps remain as 'the great unread' if only close reading was regarded as scholarly relevant. In this case, instead of humans, it is necessary to use machines as readers. If the big data is in machine-readable format, it can be organized through computational methods. But is this reading at all? If we define reading as needing a human consciousness that can

continuously compare what is read with what has been previously adopted and speculate how what is read might develop in the future, this perhaps is an exercise that a machine cannot handle. But, to be sure, artificial intelligence can be enhanced so that every newly identified character, word and sentence is compared with previous expressions in the text, and not only within one text but in the context of thousands or tens of thousands of earlier texts.

The challenge of big data

For the digital historian, distant reading becomes an issue especially when assessing big data sets, often corpora of texts. Reading and interpretation of small amounts of texts, or even a singular text, is important, but it is still necessary to have a wider discussion on big data. Digital history could not have developed without large corpora of information. The discourse on big data began in the 1960s and 1970s with the expansion of electronic media and computerization. The *Oxford Dictionary of English* defines the concept as 'data of a very large size, typically to the extent that its manipulation and management present significant logistical challenges; (also) the branch of computing involving such data'.[31] Interestingly, the *Oxford Dictionary of English* cites an early comment by a historian on big data that reveals the scepticism towards computer-aided history circa 1980.[32] The comment comes from 'The old new social history and the new old social history', an essay presented by the eminent American political scientist and historian Charles Tilly at the University of Michigan in October 1980. This reference shows that historians have always been interested in drawing on large accumulations of data. In his text, Tilly quotes at length his British colleague Lawrence Stone's critique of cliometrics, focusing particularly on the work of Robert Fogel and Stanley Engerman, who had used computers for social historical analysis.[33] According to Stone, the cliometricians 'specialize[d] in the assembling of vast quantities of data by teams of assistants, the use of the electronic computer to process it all, and the application of highly sophisticated mathematical procedures to the results obtained'.[34] There

were severe problems, however, which were echoed by Tilly, who argued that 'coding loses crucial details' and that 'mathematical results are incomprehensible to the historians they are meant to persuade'. Furthermore, 'the storage of evidence on computer tapes blocks the verification of conclusions by other historians'.[35] Tilly continues: 'None of the big questions has actually yielded to the bludgeoning of the big-data people.' He concludes with Stone's words that 'in general the sophistication of the methodology has tended to exceed the reliability of the data, while the usefulness of the results seems – up to a point – to be in inverse correlation to the mathematical complexity of the methodology and the grandiose scale of data-collection'.[36]

This reference in the *Oxford Dictionary of English* portrays an ambivalent vision of 'big data' in the late 1970s and early 1980s. It is important to remember that at that time, in addition to Fogel and Engerman's work, there were other computer-assisted projects; in Britain, for example, Edward A. Wrigley and Roger Schofield's study on the population history of England from the sixteenth to the nineteenth century,[37] and in Estonia, Juhan Kahk's several studies on rural population.[38] Computer-assisted history was particularly characterized as a business for social historians. The reference in the *Oxford Dictionary of English* puts forward a critique by two renowned social historians whose criticisms have not yet been overcome. Today, doubts regarding the interdisciplinary nature of digital history remain. How does one ensure that everything is processed correctly? How does one ensure source criticism has been taken seriously? How does one present results in a manner that is understandable and relevant to both historians and computer scientists? And how does one give future generations the possibility of verifying the results?

Since the days of Tilly and Stone's critique, much has happened. There are much more data available, of course, and not only in the archives, since massive data flows are produced and accumulated continuously, for example in Internet servers or individual mobile phones. There are, as Gartner's IT Glossary puts it, 'high-volume, high-velocity and/or high-variety information assets that demand cost-effective, innovative forms of information processing that

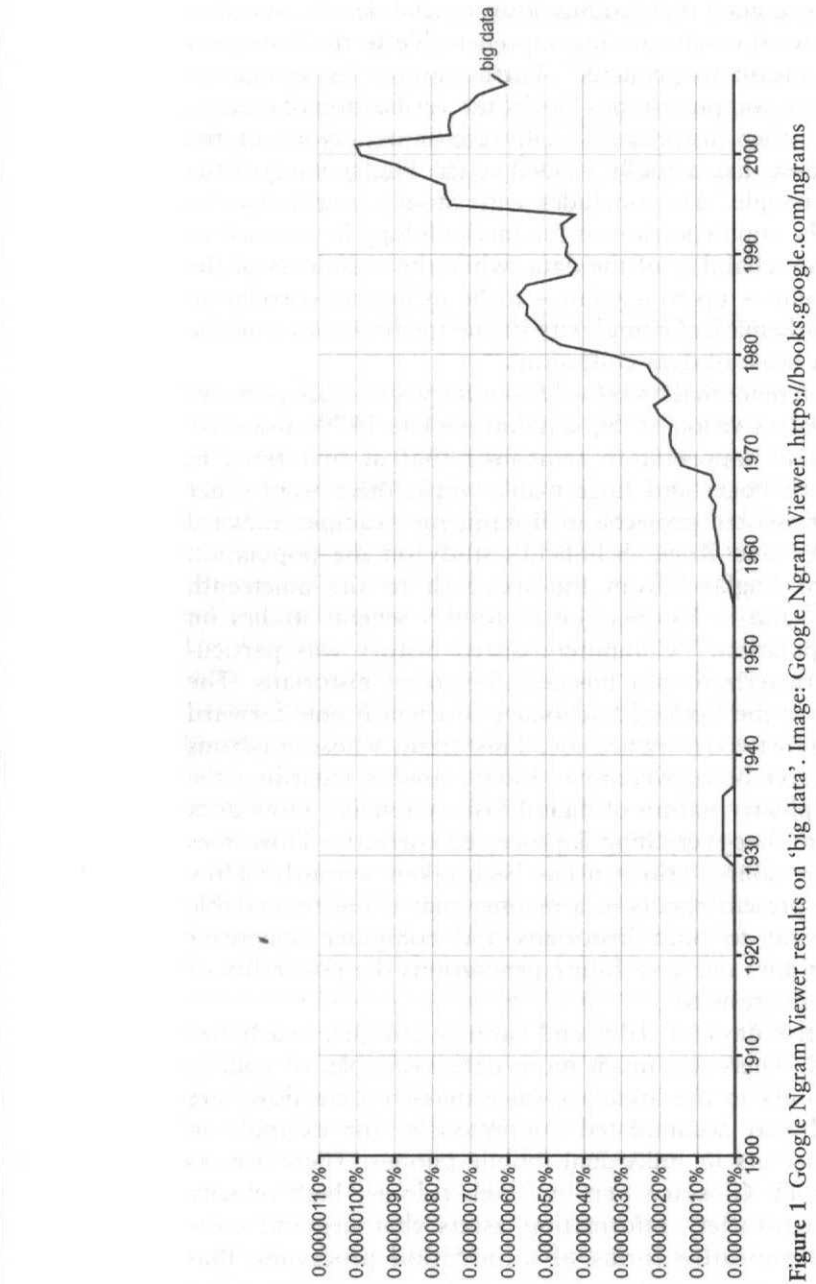


Figure 1 Google Ngram Viewer results on 'big data'. Image: Google Ngram Viewer. https://books.google.com/ngrams

enable enhanced insight, decision making, and process automation'.[39] Big data is everybody's concern. Rajan Saxena describes it as 'structured and unstructured data collected from various sources, such as social media text, cell phone locations, web browsing and search engines, call records, radio frequency identifications, maps, traffic data and internal company records'.[40] Of course, this list should include also all the historical records digitized in recent decades.

The term 'big data', as described above, already has a long history. We can illustrate this by drawing on big data itself. In 2010, Google Lab released its Google Ngram Viewer, developed by Jon Orwant and Will Brockman, which allows searches in the company's vast digitized collection of books. At the time of its opening, the tool covered 5.2 million books published between 1500 and 2008 in American English, British English, French, German, Spanish, Russian and Chinese. If the whole English corpus is searched with the character string 'big data', the Viewer gives the results shown in figure 1.

This curve seems to suggest that the use of the phrase 'big data' in the Google Books corpus grew rapidly towards the end of the twentieth century.[41] The last year in this visualization is 2008: 'big data' is still strongly present, although not as much as right after the year 2000. The reliability of these results should be met with some reservations since there are cases where the word 'big' is actually the last word of one sentence and 'data' is the first of the next, and there are clearly cases were the metadata is incorrect; for example, the year of the first edition is given while the reference to 'big data' is from a later edition. Still, in this case, big data itself gives a rough idea of the evolution of the term 'big data' in English literature.

The question of the 'bigness' of big data often arises. In their *Exploring Big Historical Data: The Historian's Macroscope*, Shawn Graham, Ian Milligan and Scott Weingart make an important point about how relative the idea of big actually is. 'Big' has different dimensions and proportions depending on the discipline: '[B]ig data for literature scholars might mean a hundred novels ("the great unread"), for historians it might mean an entire array of nineteenth-century shipping rosters, and for archaeologists it might mean every bit of

data generated by several seasons of field survey and several seasons of excavation and study.'[42] Notwithstanding their comment that 'the great unread' constitutes only a hundred novels in comparison to those thousands of volumes that Moretti refers to, the researchers are right to note that there is no quantitative criterion or limit to the size of data. Everything depends on the research setting and the questions asked. In fact, everything depends more on the questions than the discipline; for example, historians' interests are not restricted to shipping rosters but can involve novels and excavation reports as well. The size of the material must be estimated separately for each case. Furthermore, the term 'small data' is used to stress that not everything has to be big and that digital methods can be applied to any kind of data.

As the previous discussion shows, big data can be any kind of data, from textual material to visual sources, from social media content to location information. Most commonly, the big data that researchers currently use in the digital humanities and digital history are vast collections of either digitized or born-digital texts. This tendency has been so strong and so deeply rooted that there has been growing criticism of the field's over-textual approach and arguments that visual sources should be taken more seriously. 'Textual analysis, in all shapes and forms, has come to dominate the field', remarked Melvin Wevers and Thomas Smits in a recent article. 'This emphasis', they concluded, 'stems from the availability of technology to study digitized text. Optical character recognition allows researchers to use keywords to search and analyze digitized texts.'[43] While this conclusion clearly hits the point, it is as obvious that textual material has been and will be of central value for any kind of research in the humanities.

Texts for historical research

In historical studies, there is a long tradition of analysing the textual remains of the past, both manuscripts and printed texts. When academic historical research really started in the nineteenth century, epitomized by such historians as Leopold von Ranke in Germany, textual sources were highly valued,

and not only textual but archival sources in particular.[44] According to that view, there would be no history if recorded sources had not been available. The French historians Charles-Victor Langlois and Charles Seignobos expressed this principle famously in their 1898 *Introduction aux études historiques*, translated into English in 1904 as *Introduction to the Study of History*: 'No document, no history.'[45]

In fact, the early days of academic history produced a kind of hierarchy of textuality that has had an impact on the concept of history: archival official documents were ranked above private correspondence, and handwritten texts were ranked above printed and more popular publications. In that sense, textuality was never regarded as one genre but as an overarching category for many kinds of cultural products. Fuelled by their passionate devotion to texts, historians developed their skills in source criticism and hermeneutic interpretation. Towards the end of the nineteenth century, however, and increasingly in the early twentieth century, historians were criticized for being too dependent on written and printed sources and failing to acknowledge other traces of the past. Such pioneers as the Dutch historian Johan Huizinga emphasized visual sources in his 1919 *Waning of the Middle Ages*, which starts with a look at the enigmatic paintings by the Van Eyck brothers.[46] To be sure, the Middle Ages could hardly be understood without visual material. From the 1920s onwards, historians from the French Annales School, for example, suggested that any kind of cultural product could be a source of information for a historian, not only written and visual sources but also oral tradition, material remains, buildings, statistics and audio-visual sources.[47] From this perspective, the discussion in the digital humanities seems sometimes like a return to those similar views that were debated 100 years ago: texts, especially printed ones, are easy to access, and there is already a methodology for their successful employment in research. But this is, of course, only partially true, since the context is different today. One of the reasons for the emphasis on textual big data today is the fact that historical linguistics has had a long tradition of studying digitized corpora; researchers of this field were doing so long before such corpora were found by historians and other humanities researchers.

Today, printed textual sources are intensively used, since large digitization projects of the sort discussed in the previous chapter have reached the point where they not only offer a rich basis for scholarly inquiry but provide possibilities for linking these corpora together. Many text collections have been produced by national libraries or through private initiatives, and therefore transnational or transregional research problems require substantial amounts of preparatory work to make larger research settings possible.

Newspapers are probably the most common form of digitized source material for historians today. It might be argued that, as early as the nineteenth century, newspapers were a form of big data *avant la lettre*. Since the 1830s, high-speed presses have allowed the inexpensive printing of tens of thousands of newspapers in a day. The English politician and author Sir Edward Bulwer Lytton described newspapers as 'the chronicle of civilization, the common reservoir into which every stream pours its living waters, and at which every man may come and drink'.[48] There was newspaper publishing, of course, since the seventeenth century, and these early papers are important for digital historians, too, but since the beginning of the nineteenth century, the periodical press formed a network that reached all corners of the world. Because of that, they offer a basis for large-scale analyses, a basis that transgresses continental borders.

Over the last fifteen or twenty years, national libraries have digitized their newspaper collections. If we add up the digitized papers of, for example, Australia, Finland, Germany, Mexico, New Zealand, the Netherlands, the United Kingdom and the United States, we get over 100 million pages of content. There is no search engine that would enable the inquiry of all these collections with only one search, and researchers must consult each of these repositories separately.

There are many curated lists of historical newspaper archives that help researchers navigate globally and find relevant collections. One of the most comprehensive lists can be found on Wikipedia, which includes digitized collections from countries like Argentina, Brazil, Canada, Equatorial Guinea, Iceland, Japan, Russia and Slovenia.[49] Self-evidently, there is a lot of variation in these collections, both in terms of accessibility and search options. Some collections are behind

pay walls, such as the *Japan Times Digital Archive*[50] and *Frankfurter Allgemeine Zeitung*,[51] while some are free for all visitors, like the Digital Newspaper Archives of the National Library of Greece[52] and the Hemeroteca Digital at the National Library of Spain.[53] Another issue is searchability. It seems that some of the projects have been launched under circumstances in which searchability was not a priority; rather, the idea was to make the publications available online for readers who cannot access the library itself. For example, the newspaper *La Guinea Española* can be found online as PDFs from 1903 to 1969.[54] Each issue is a separate file that has been saved only as an image, and the text has not been optically recognized and so is unsearchable. The resource is valuable for scholars of Equatorial Africa, but to enable the use of digital methods, the user should first download all PDFs, extract the texts with OCR software and perhaps pursue other forms of pre-processing.

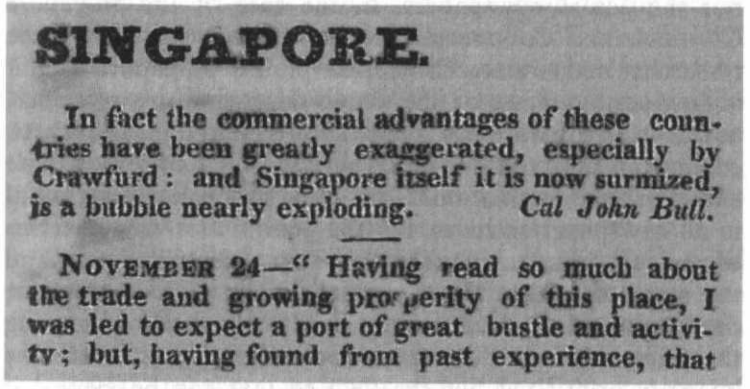
Andrew Prescott pointed out that in many digitization projects, the original aim was not to produce a searchable text, a point that applies to many of today's most used collections.[55] Prescott was referring to the Burney Collection, a well-known British newspaper resource. This project started after the British Library bought a microfilm digitizer in 1992: 'It was an experimental project to improve the way in which the Library made its microfilm surrogates available. There was no assumption that digital images would replace microfilm, since microfilm, which tests showed would survive one thousand years in the right conditions, was considered a more stable preservation medium.'[56] Today, the Burney Collection is a searchable service provided by Gale Cengage, but at the start of the project in the 1990s, it was not clear how technology would develop in the future and what would be the best option for preservation. In the 1990s, there was also a Scandinavian project called TIDEN, which included partners from Denmark, Finland, Norway and Sweden and marked the start of the National Library of Finland's digital collection. Already in 1990, the Helsinki University Library established the Centre for Microfilming and Conservation in the small, eastern town of Mikkeli with the idea of creating a comprehensive microfilm collection of Finnish newspapers and journals. It soon became obvious that digitization would

be more fruitful for the development of research infrastructures, and the TIDEN project offered the possibility of concentrating on the exploration of digitization problems as well as the production of actual digital copies, mostly on the basis of previously produced microfilms.[57] This is the reason why the present collection of digitized newspapers in Finland is almost completely based on microfilm, as are many other collections in the world. Today, it would be laborious, and expensive, to scan everything again. This story has led to several consequences, of which the question of searchability is only one. Another issue is the quality of the digitization itself. Somewhat poor quality did not matter in the beginning, since the only motive was to give easier access for users who were earlier forced to roll microfilms. Soon, however, the development of search functions appeared, but the quality of the scan limited the work. This is a reminder of the importance of source criticism and the need to consider how the formation of big data collections influences the condition under which they are used as historical evidence.

The search interfaces of digitized text collections are based on optical character recognition. The methods to capture text from scanned images have been developed since the late nineteenth century, but they became practical research tools from the 1970s onwards.[58] One of the most successful OCR programs is the ABBYY FineReader, released in 1993. The latest version of the software can recognize 192 different languages.[59] The development has been tremendous, but OCR still creates a problem for historical research, since the further back in time we go, the more difficult character recognition seems to be, because of the instability of font shapes, the differences in original printing materials and the multi-layered scanning history of a given source, including that so much has been digitized from microfilm. Tim Hitchcock has pointed out that OCR 'does not produce reliable results when applied to early modern publications, the cheap and quotidian print of the nineteenth century, or any form of complex formatting including tables, lists and advertisements'.[60] Furthermore, the quality does not vary historically, in the sense that older material would be automatically poorer than modern material. For example, in the late nineteenth century there were often six, seven or

even eight columns per page, and although the paper size increased, the font size kept diminishing, which again makes it difficult for the OCR program to correctly recognize the text. Typesets also present a problem: current software can more easily recognize modern fonts than the older Gothic typeset that was popular before the twentieth century. The OCR methods are, however, becoming better for both historical and non-European sources, and new solutions will soon be in the toolbox. For example, The Early Modern OCR Project (eMOP) at the Texas A&M University offers an excellent example of the imaginative ways of combining open source OCR software and book history to improve the accuracy of OCR for early modern printed texts.[61]

All this means that digitized text repositories contain a lot of digital noise, misread characters and OCR errors, resulting in the problem that interface searches can never find all the relevant information.[62] This can be illustrated with an example from the *Singapore Chronicle and Commercial Register*, which is available online.[63] On the website of the National Library of Singapore, the reader can see the visual content of the paper, article by article. On 15 February 1827, the *Singapore Chronicle and Commercial Register* published a news item on the commercial significance of Singapore. The text is actually copied from the Calcuttan newspaper *John Bull*, dated 24 November 1826, and the source is mentioned



SINGAPORE.

In fact the commercial advantages of these countries have been greatly exaggerated, especially by Crawfurd: and Singapore itself it is now surmized, is a bubble nearly exploding. ***Cal John Bull.***

NOVEMBER 24—" Having read so much about the trade and growing prosperity of this place, I was led to expect a port of great bustle and activity; but, having found from past experience, that

Figure 2 The *Singapore Chronicle and Commercial Register*, 15 February 1827. The digitization has been made from a microfilm, provided by the Microform Imaging Ltd. Image: Microform Imaging.

both in the beginning and at the end. The text starts as shown in figure 2.

For a human reader, the text in figure 2 is quite easy to read. The equivalent OCRed text for this clip, however, runs as follows: 'SINGAPORE. is™ bubble nearly Cal John Bull. November 24-" Hiving r .f d .""A" the trade and growing prospenty of Uiis place l -was led to expect a port or great bustle ana aetty ty; but, having found from past ex P^'' e,f C, b 5'.

As we can see, the first four lines from *John Bull* are only partially recognized by the OCR software. The first three lines are missing completely, as well as the word 'exploding' from the fourth line. There were already spelling errors in the original newspaper, for example 'Crawfurd', instead of 'Crawford', but the OCR software produced even more errors, like 'Hiving' instead of 'Having' and 'prospenty' instead of 'prosperity'. There are even characters that do not belong to the nineteenth-century text at all, for example the superscript abbreviation ™, meaning trademark, and a series of characters that do not fit the contemporary language model either, such as 'P^&#39'. These kinds of algorithmic interpretations of historical texts are familiar to all users of digitized newspapers. These kinds of noisy fragments can be found in all newspaper repositories. From a source-critical perspective, everybody who consults these repositories must consider the fact that all queries address the OCRed text, not the actual newspapers. In the case of the *Singapore Chronicle and Commercial Register*, this means that if the researcher makes a search with the phrase 'Singapore itself it is now surmised', he or she would never find this reference, as the OCR software has totally ignored that line. The search engine can only retrieve hits where the OCRed text is close enough to the original one. This same problem can be found in all newspaper archives, but the possibilities vary in terms of reviewing the relationship between the OCRed text and the scanned image. There are collections that do not give the text at all, as if the search had really been addressing the scanned image. Likewise, there are services in which the image is prioritized but the OCRed text can be viewed if wanted. One of the clearest sites in this sense is Trove, the newspaper collection of the National Library of Australia,

which opens the OCRed text and the image in the browser at the same time and in separate windows.[64]

Johan Jarlbrink and Pelle Snickars have pointed out that the ways 'digitized data (as newspaper files) are created, stored, processed and formatted naturally have implications on how historical records as digital newspapers can be accessed and used, the histories that can be explored and the stories that can be (re)told'. They further emphasize that digital noise is already a substantial part of our cultural heritage and that, in addition to historical records, the collections include 'millions of misinterpreted words generated by OCR', and in fact millions of texts that have been re-edited by computational tools.[65] Ryan Cordell has also discussed the problem of 'dirty OCR' at length and suggested that instead of seeing this only as a major obstacle for research, it would be appropriate to regard digitized newspapers as 'new editions of their source texts that offer unique insights into both the historical texts they remediate and the more recent era of their remediation'.[66] To be sure, newspaper repositories include the layers of those historical processes that produced them. The challenge for the future is how to express this cumulative nature of collections in the best possible way so as to offer the user an insight into the editorial history, both handmade and algorithmic, of the sources provided. There are many recent projects that have concentrated on developing newspaper repositories, including the project *Media Monitoring of the Past: Mining 200 years of historical newspapers*, funded by the Swiss National Science Foundation,[67] and *NewsEye: A Digital Investigator for Historical Newspapers*, funded by the European Union's Horizon 2020 research and innovation programme.[68] The major problem, however, is ensuring that the tools and guidelines developed with project funding, often granted for only a few years, can best benefit libraries and archives in their long-term enhancement processes.

One of the challenges in newspaper repositories – and to be sure in all digital text collections – is that they are siloed in different libraries and archives. It is, of course, possible for a historian to consult each of these collections separately, and there are many examples of the benefits of this kind of work,[69] but still the bridging of these collections would inspire researchers to explore transnational

connections and see beyond regional silos. One of the largest enterprises to bring national repositories together has been the Europeana project, which also includes a section for newspapers, covering material from twenty European countries from 1618 to 1996.[70] This collection, too, has the problem of temporary funding, but it shows clearly the benefits of bringing resources together. Europeana allows the user to search the database using keywords and phrases, which has indeed been laborious since the metadata accompanying individual newspapers varies from country to country. Another example of the efforts to bring resources together is the project *Oceanic Exchanges: Tracing Global Information Networks in Historical Newspaper Repositories, 1840–1914*, which was led by Ryan Cordell. It brought together the researchers of digitized periodicals from six countries – Finland, Germany, Mexico, the Netherlands, the United Kingdom and the United States – to examine patterns of information flow across national and linguistic boundaries.[71] The project has also produced data descriptions for each country, including information on the history of the collections and the quality of their metadata.[72] To date, one of the project's main results has been the observation of how many differences exist in the architecture of metadata. The complex diagram produced by Melodee Beals and the project team may, however, offer an avenue for future linkages of these data collections.[73]

This discussion on text archives has only dealt with the basics of the repositories that allow searches through their interfaces. For historians, the use of search interfaces is essential, but for further digital analysis of the material it is necessary to be able to download the data, for example, as a data dump that includes the OCRed texts and metadata, often in international XML standards, such as the METS and ALTO formats.[74] In a data dump, all the files are packaged so that they can be easily transferred from one location to another. When the package is unzipped, it usually includes folders of volumes and issues of newspapers. Once these packages are downloaded, the researcher can employ more advanced data mining tools and really gain from the advantages of big data. An excellent site from the perspective of testing with data dumps is provided by the National Library

of Luxembourg, since, in addition to large data sets of several hundreds of gigabytes, it also includes test packages. The smallest dump is only 250 MB, and it covers five days in 1868 of the Luxembourgian newspaper *D'Wäschfra*. Once the dump is downloaded and unzipped, the user finds five folders, one for each day. Each folder includes the issue both as a PDF file and as an XML. It also includes four folders. The image folder includes every page as a high-resolution TIFF image, the PDF folder includes individual pages as PDFs, the text folder has each page separately as an XML and the thumbnails folder is comprised of smaller images of every page as a JPG file.[75] Therefore, as a service, the dump includes every issue as a PDF or XML file, but it is possible to draw on separate page files, depending on the researcher's interests.

The previous examples of textual big data are from the periodical press, but there are indeed many other text collections that can be employed. Some of them were presented in chapter 1, such as the multilingual bookshelf provided by Project Gutenberg, and these examples can be supplemented with other online e-book collections from around the world, born-digital content harvested from the Internet, oral historical transcriptions that have been digitized, and many other sources. An intriguing challenge would be to combine these separate collections in the future and pose research questions that go beyond the boundaries of different textual genres. This still has yet to be done. Furthermore, the question of textual data does not today refer only to digitized printed sources or born-digital content; handwritten material is also rising quickly on the agendas of digital historians. Handwritten text recognition (HTR) has developed rapidly over the last few years, and the results are improving. In some cases, HTR gives more accurate results than the OCR of printed sources. There are several programs available in the field. One of the most stimulating projects has been Transkribus, which is hosted by the University of Innsbruck and funded by the European Commission as part of the H2020 Project READ in 2016–2019. Transkribus gives the user the possibility of teaching the program by giving it samples from a set of historical documents. This information again helps developers enhance the software further, since

more training material is constantly provided.[76] Amsterdam City Archives has used Transkribus in its HTR projects. At its best, the error rate in HTR has been only 6 per cent.[77]

Distant reading techniques

As described above, there are numerous textual corpora that digital historians can employ. Let me return to the question of reading. What does reading mean, and what would it entail, if the researcher has to adopt millions of words, or even more. How can interpretation be realized in such a case?

Reading has always involved interpretation and this is the case also in distant reading, which is not against human interpretations. Rather, it can merely be argued that distant reading divides reading into different processes: the recognition of text, its organization and its structural analysis, for example, can be done by a machine reader and the interpretation of this process by a human reader, specifically, the researcher. The process of human interpretation can also be computer-aided or run by artificial intelligence.

In the present-day digital humanities, and digital history, distant reading does not refer to any single way of reading or any coherent set of methodologies. Many solutions for organizing big data have been developed during recent decades. They have not been invented as responses to the idea of distant reading, since some of them have longer histories, but they have clearly gained more weight in the toolboxes used by humanities researchers as ways of coming to terms with large data collections.

Topic modelling is a statistical method for studying underlying structures in a series of documents.[78] 'Topic', in this case, refers to a group of words that tends to occur in the data. Modelling these topics is an open-ended process, as the amount of topics is ultimately a result of how the researcher seeds the algorithm.[79] The most common statistical model is latent Dirichlet allocation (LDA), which was developed in 2002 by David Blei, Andrew Ng and Michael I. Jordan.[80] There are also easy-to-use software programs, like Mallet, for topic modelling.[81] The main idea is that the more data researchers have, the more useful topic-modelling tools are

for organizing that data, finding which words tend to cluster together and identifying topics. Topic modelling is useful for unstructured text in which there are no annotations that would help the computer interpret semantic meaning. It is important not to mix these topics with themes or to see the two as identical. For example, a text might have a fixed number of themes that can be observed through careful close reading. In topic modelling, however, the algorithm finds as many topics as the researcher has given in the settings of the program. If the number is too high, the topics might be dispersed, which makes their semantic interpretation difficult. In practice, the researcher must try to find the most fruitful number of topics for the analysis.[82]

There are many examples of the use of topic modelling in historical research. In 2011, for example, Tze-I Yang, Andrew J. Torget and Rada Mihalcea experimented with topic modelling by using historical newspapers as a data set. They analysed papers published in Texas from 1829 to 2008, reviewing over 230,000 pages, which is already a big collection of data. They divided the material into four subgroups for testing topic modelling and pre-processed the data before analysis. In this process, so-called stop words also had to be identified. These stop words are commonly used words, like pronouns, that must be removed before running the algorithm so as to make the results as fruitful as possible. Finally, the researchers were able to extract historically meaningful topics from a large amount of data on Texan history, but they also confronted the problem of noise in the optical character recognition.[83] If the OCR is bad, it means that there are so many broken or misspelled words that the results become confused. Despite this, the study, and many other case studies,[84] revealed the potential of statistical analyses such as topic modelling in helping to organize data and inject semantic meaning.

Topic modelling is only one example of distant reading. Moretti himself emphasized visual techniques, or as he put it, 'graphs, maps and trees'. With graphs, he wanted to visualize the changes in the genre of historical novels with maps to illustrate the geographical aspects of fiction and with trees to show how different types of stories could be classified.[85] In 2015, in their effort to classify different close and distant

reading techniques in the digital humanities, Jänicke et al. also identified, in addition to 'graphs, maps and trees', other categories of distant reading, such as structure overviews, heat maps, tag clouds and timelines. They also tried to identify strategies in combining close and distant reading, the ways in which users of big data could return to small data and even individual documents in their effort to interpret the results.[86]

The realm of distant reading is further illustrated on The Programming Historian, a website that has continuously aimed at encouraging historians to learn computational methods themselves.[87] The 'novice-friendly' site consists of peer-reviewed tutorials written in English, French and Spanish. As of May 2020, the English site already included eighty-two lessons on various themes, including the use of Mallet, but also more advanced issues, such as using R-language for data wrangling and management or Python for stylometrics, which is the analysis of style. Under the heading 'Distant Reading', the website has eleven tutorials, and these themes show what methods are, on a preliminary level, associated with distant reading techniques. In addition to data wrangling, stylometrics and topic modelling, these lessons include, for example, corpus analysis with AntConc software and an introduction to a database called MySQL to store and filter data. The lessons also include sentiment analysis as a method for quantifying the degree of positivity or negativity within a text or corpus.[88] The list of distant reading strategies could be supplemented with many more. Moretti himself was particularly interested in networks, and he visualized, for example, character networks of Shakespeare's plays.[89] The Programming Historian offers several lessons on networks for self-learning.[90] Shawn Graham, Ian Milligan and Scott Weingart have also given an introduction to network analysis, in addition to many other distant reading techniques, in their 2016 guidebook *Exploring Big Historical Data: The Historian's Macroscope*.[91]

One of the most vibrant areas of distant reading is the study of text reuse, which can include exact quotations but can also cover intentional or unintentional borrowing and paraphrasing.[92] Repeated and copied passages can, of course, be identified through close reading, which theologists, historians, literary scholars and other researchers in the humanities

have done for centuries. The larger the amount of text, the more beneficial the use of digital methodologies is. Today, it is possible to detect similarities in vast corpora, such as the data dumps of millions of newspaper pages. Here, distant reading helps to algorithmically recognize relations within the corpora, which again allows the researcher to close read, for example, chains of repeated media content. Algorithmic solutions can be based on detecting similar character strings, with tolerance to OCR errors and other forms of noise, or locating sequences of words and how words occur together.

There have been several fascinating projects on text reuse in recent years. Lincoln Mullen analysed biblical quotations in American newspapers;[93] Marco Büchler, Gregory Crane, Maria Moritz and Alison Babeu studied how Homeric quotations appeared in ancient texts;[94] and the project *Knowledge, Information Technology and the Arabic Book* explored text reuse in the corpus of Arabic texts from the period 700–1500.[95] Perhaps the most ambitious endeavour has been the Viral Texts project, led by Ryan Cordell and David A. Smith, which aims to analyse text reuse in the American press of the nineteenth century.[96] The project has been enlarged to a global scale and brought together researchers from Finland, Germany, Mexico, the Netherlands, the United Kingdom and the United States to explore transnational and transcontinental flows of information.[97]

The culture of copying has many historical ramifications and can shed interesting light on the repetitive nature of the past. This can only become visible from a distance. In a text reuse project from Finland, it was possible to mine all the published issues of newspapers and journals from the birth of the local newspaper business until 1920 – in sum, 5.1 million pages. In the end, 13.8 million clusters of reused texts were found. Most of the cases of content-sharing happened within one year, but over two million clusters lasted longer. Astonishingly, there were also strikingly slow processes of repetition; the longest reuse cases were almost as long as the time span of the project.[98] This long-term perspective shows the benefits of computational methods.

Distant reading, or computer-assisted close reading, means that historians must engage with technology in one way or another. The Programming Historian and many

recent guidebooks have paved the way for this by spreading knowledge and knowhow on useful methodologies and the necessary computational diagrams behind them.[99]

As already pointed out in this chapter, the dominating role of textual sources in the digital humanities faces growing criticism, much as it does in digital history. Before having a closer look at visual and audio-visual materials in the next chapter, it is worth recalling that the supposed duality between textuality and visuality is quite misleading and should not be taken too strictly. Texts include images and vice versa. Today, Internet texts often include both sound and visuals. And, moreover, both texts and images can be considered as material remains of the past. Text is not an abstract order of characters but is always, to draw on Bruno Latour's idea, inscribed on material objects, on stone or marble, on paper or hard drives.[100] Texts are material entities that have dimensions: printed texts manifest themselves, for example, as folios and octavos, as broadsheets and tabloids. When considering texts as big data, metatext must seriously be taken into account, since it offers access to the material dimensions of the texts at hand. Metadata has recently been successfully used in the study of book history, in which it is vital to comprehend how books circulated as concrete artefacts in the past.[101] In the future, these material ramifications might be avenues for bringing the quantifiable data of texts more closely together with the study of their contents and, thus, their impacts on the imaginations of readers.

3

Mapping and Viewing History

In the early stages of digital history, in the 1990s, maps were of particular interest, perhaps more so than the textual materials discussed in the previous chapter, because large-scale digitization projects had only recently begun. At the University of Virginia in 1993, William G. Thomas and Edward L. Ayers started a project on the US Civil War, entitled *The Valley of the Shadow*, that brought together the experiences of Confederate and Union soldiers. It published historical documents online, including letters, diaries, images, newspapers, church records, census records, tax records, statistics – and maps. The pre-war material, for example, included maps pertaining to geography, infrastructure, agriculture, politics, religion and slavery in the selected counties (Augusta in Virginia and Franklin in Pennsylvania). These were not merely digitized pre-existing visualizations but also an integral part of the project, produced for online use by the project members.[1]

Visual sources have been at the heart of digital history from the start, and maps have been an essential ingredient in visualizing archival data and establishing points for comparison. The following chapter continues by discussing maps further as sources and historical interpretations, fuelled by the so-called spatial turn. The chapter proceeds by selecting three other themes under discussion, the increased role of visual sources, including the idea of distant viewing,